A Comparative Study of Convolutional and Transformer Architectures Under Adversarial Perturbations: Gradient Characteristics and Transferability Analysis

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***Abstract*—Adversarial robustness of deep neural networks remains a critical concern for safety-critical applications. While Vision Transformers (ViT) have achieved remarkable success on clean data, their vulnerability to adversarial attacks compared to Convolutional Neural Networks (CNNs) requires deeper investigation. This paper presents a comprehensive comparative analysis examining *why* these architectures exhibit different adversarial behaviors rather than merely comparing robustness scores. Using CIFAR-10 with adversarially trained ResNet-18 and ViT-Tiny models evaluated under AutoAttack ($\epsilon = 8/255$), we demonstrate four key findings: (1) CNNs achieve 36.0% robust accuracy compared to 32.4% for ViTs, a gap that persists across attack types; (2) Transfer attacks exhibit significant asymmetry—CNN-generated adversarial examples transfer to ViT at 47.5% success rate while ViT-to-CNN transfers succeed at only 33.5%; (3) Gradient analysis reveals ViT gradients are 2.2$\times$ more aligned across samples and CNN gradients are 4.5$\times$ more sparse, explaining the transfer asymmetry; (4) Feature degradation in ViT increases progressively through layers, with late layers showing 7.86% norm reduction and cosine similarity dropping from 0.997 to 0.917. These findings provide mechanistic insights into architectural vulnerability differences and suggest design principles for robust hybrid systems.**

***Index Terms*—Adversarial robustness, Vision Transformer, Convolutional Neural Networks, Transfer attacks, Deep learning security**

I. Introduction

Deep neural networks have achieved remarkable success across various computer vision tasks, including image classification, object detection, and semantic segmentation. However, these models are vulnerable to adversarial examples—carefully crafted perturbations that are imperceptible to humans but can cause misclassification with high confidence [1], [2]. This vulnerability poses significant security concerns for safety-critical applications such as autonomous driving, medical diagnosis, and security systems.

Two dominant architectural paradigms have emerged in computer vision: Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). CNNs leverage local connectivity and weight sharing through convolutional operations, capturing hierarchical spatial features effectively [3]. In contrast, ViTs apply the self-attention mechanism from natural language processing to image patches, enabling global context modeling from early layers [4]. While both architectures achieve competitive performance on clean data, their behavior under adversarial attacks exhibits fundamental differences that are not yet fully understood.

Previous research has extensively studied adversarial robustness for CNNs, leading to effective defense mechanisms such as adversarial training [5] and TRADES [6]. However, the comparative analysis of CNN and ViT robustness remains limited, with existing studies often focusing on *whether* one architecture is more robust than another, rather than *why* they exhibit different behaviors. Understanding the mechanistic factors underlying these differences is crucial for developing more secure deep learning systems.

In this work, we move beyond simple robustness comparisons to investigate the underlying mechanisms that cause CNNs and ViTs to respond differently to adversarial perturbations. Our investigation centers on four interconnected questions. First, we examine why adversarial perturbations transfer asymmetrically between architectures—CNN-generated examples transfer to ViTs with significantly higher success than the reverse direction. This asymmetry motivates our second question: whether gradient characteristics can explain the observed robustness differences. Third, we trace how adversarial perturbations propagate through Vision Transformer layers, seeking to understand the feature-level dynamics that lead to misclassification. Finally, we ask what design principles emerge from these mechanistic insights for building robust hybrid systems that leverage the complementary strengths of both architectures.

This paper makes several contributions to understanding architectural differences in adversarial vulnerability. We provide a comprehensive robustness evaluation using AutoAttack [7], the gold standard benchmark, comparing adversarially trained ResNet-18 and ViT-Tiny under $\epsilon = 8/255$ L_∞ perturbations. Our transfer attack analysis reveals and explains the significant asymmetry in cross-architecture transferability, demonstrating that this asymmetry stems from fundamental differences in how the architectures compute gradients. Specifically, we identify that CNN gradients exhibit higher sparsity while ViT gradients show greater cross-sample alignment—properties that directly determine perturbation transferability. Furthermore, we present the first systematic layer-wise analysis of how adversarial perturbations affect ViT feature representations, revealing progressive semantic degradation that intensifies with network depth. To support reproducible research, we release our complete experimental pipeline, trained models, and analysis tools.[1]

We begin by reviewing related work on adversarial attacks, defenses, and architecture-specific robustness studies in Section II. Section III describes our experimental methodology, and Section IV presents results across four complementary analyses: robustness comparison, transfer attacks, gradient characteristics, and feature degradation. We discuss the implications of our findings in Section V and conclude in Section VI.

II. Related Work

A. Adversarial Attacks

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f(x), y)) \tag{1}$$

where ϵ bounds the perturbation magnitude under the L_∞ norm.

Projected Gradient Descent (PGD) [5] extends FGSM through iterative optimization:

$$x^{t+1} = \Pi_{B_\epsilon(x)} \left(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}(f(x^t), y)) \right) \tag{2}$$

where $\Pi_{B_\epsilon(x)}$ projects onto the ϵ-ball around the original input and α is the step size.

The Carlini & Wagner (C&W) attack [8] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [7] combines multiple attack strategies (APGD-CE, APGD-DLR, FAB, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

B. Defense Mechanisms

Adversarial training (AT) [5] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_\theta \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\|\delta\| \leq \epsilon} \mathcal{L}(f_\theta(x + \delta), y) \right] \tag{3}$$

TRADES [6] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\text{TRADES}} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \tag{4}$$

where β controls the trade-off parameter.

MART [9] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [10], certified defenses via randomized smoothing [11], and ensemble methods [12]. Recent advances in certified robustness include Adaptive Randomized Smoothing [13], which improves certification radii through multi-step defenses, and diffusion-based purification methods such as ADBM [14] that achieve competitive robustness against unseen threats.

C. CNN Robustness

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [5] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [15] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [5], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the importance of skip connections [3] and batch normalization placement [16].

D. Vision Transformer Robustness

Vision Transformers [4] process images as sequences of patches using self-attention. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [17]. However, subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [18].

Shao et al. [19] demonstrated that adversarial training can improve ViT robustness, though challenges remain due to training instability. The relationship between patch size, attention mechanisms, and adversarial vulnerability has been explored [20]. More recently, Jain et al. [21] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100. Wei et al. [22] further investigated ViT vulnerabilities through gradient normalization and high-frequency adaptation. Despite these advances, comprehensive mechanistic comparisons between CNN and ViT robustness under unified evaluation frameworks remain limited.

E. Transfer Attacks

Adversarial examples often transfer between models, enabling black-box attacks [23]. The transferability depends on factors including model architecture, training data, and attack methods [24].

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Naseer et al. [25] explored transfer attacks from CNNs to ViTs, finding reduced transferability compared to within-architecture transfers. Recent work has proposed specialized methods for cross-architecture attacks: Token Gradient Regularization (TGR) [26] crafts transferable adversarial samples by exploiting ViT's attention components, while Momentum Integrated Gradients (MIG) [27] achieves improved transferability across both ViTs and CNNs. Wang et al. [28] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive

[1] Code and models available at: https://github.com/[anonymous]/cnn-vit-adversarial-analysis

biases—CNNs preferring high-frequency information while ViTs favor low-frequency patterns—significantly impact transfer success. Understanding these transfer patterns is crucial for developing robust ensemble systems and evaluating real-world attack scenarios.

F. Research Gap

While individual studies have examined CNN and ViT robustness separately, existing work primarily focuses on *comparing* robustness scores rather than *explaining* why architectures behave differently. Existing comparative studies typically report robustness metrics without mechanistic analysis, use inconsistent perturbation budgets or attack configurations across architectures, compare models with significantly different parameter counts, and focus on single aspects such as robustness or transferability in isolation.

Our work addresses these limitations by providing mechanistic explanations for observed differences. We investigate the gradient characteristics underlying transfer asymmetry, analyze layer-wise feature degradation in ViTs under adversarial perturbations, and connect these findings to design principles for robust systems. We employ the AutoAttack benchmark for reliable evaluation and demonstrate that gradient sparsity and alignment directly explain the observed behavioral differences.

III. METHODOLOGY

A. Threat Model

We consider the standard L_∞ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \tag{5}$$

Following established conventions for CIFAR-10 [15], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional experiments at $\epsilon \in \{2/255, 4/255, 16/255\}$ for epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

B. Model Architectures

1) CNN Models: **ResNet-18** [3]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial 3×3 convolution instead of 7×7).

WideResNet-28-10 [29]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use pretrained robust weights from RobustBench [15].

2) Vision Transformer Models: **ViT-Tiny**: We use the ViT-Tiny architecture from the `timm` library [30], adapted for CIFAR-10 by resizing inputs to 224×224. The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. This configuration enables direct comparison with pretrained ImageNet backbones while maintaining similar scale to ResNet-18.

C. Attack Methods

1) FGSM: Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \tag{6}$$

2) PGD-20: Iterative attack with 20 steps, step size $\alpha = 2/255$, and random initialization:

$$x^{t+1} = \Pi_{B_\epsilon(x)} \left(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}_{CE}(f(x^t), y)) \right) \tag{7}$$

with $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$.

3) AutoAttack: We use AutoAttack [7], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-DLR (Auto-PGD with difference of logits ratio loss), FAB (Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble represents the gold standard for robustness evaluation.

D. Defense Methods

1) Adversarial Training (AT): We train models on PGD-generated adversarial examples:

$$\min_\theta \mathbb{E}_{(x,y)} \left[\mathcal{L}_{CE}(f_\theta(x_{adv}), y) \right] \tag{8}$$

with inner maximization using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$).

2) TRADES: Trade-off between natural accuracy and robustness:

$$\mathcal{L} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \tag{9}$$

We use $\beta = 6.0$ following the original paper [6].

E. Training Protocol

1) Clean Training: For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule, weight decay 5×10^{-4}, and 200 epochs. For ViTs, we use AdamW optimizer with initial learning rate 10^{-3}, cosine annealing, weight decay 0.05, and 200 epochs.

2) Adversarial Training: Starting from pretrained clean models, we fine-tune with reduced learning rates: CNNs use SGD with LR 10^{-3} and cosine annealing for 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping. We use batch size 128 for all experiments.

F. Evaluation Metrics

1) Robustness Metrics: We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack).

2) Transfer Attack Metrics: For source model S and target model T:

$$\text{Transfer Rate} = \frac{|\{x : f_S(x_{adv}^S) \neq y \land f_T(x_{adv}^S) \neq y\}|}{|\{x : f_S(x_{adv}^S) \neq y\}|} \quad (10)$$

For transfer attack analysis, we compute success rates on the full test set (10,000 samples) to ensure statistical stability. Given the large sample size, differences exceeding 5% absolute are statistically significant at $p < 0.001$ under binomial proportion tests.

3) Gradient Analysis Metrics: We analyze gradient properties through three complementary metrics. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity quantifies the fraction of gradient components below threshold ($|g_i| < 10^{-6}$), indicating localization of sensitivity. Gradient alignment measures the average pairwise cosine similarity across samples:

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i} \mathcal{L} \cdot \nabla_{x_j} \mathcal{L}}{\|\nabla_{x_i} \mathcal{L}\| \|\nabla_{x_j} \mathcal{L}\|} \right| \quad (11)$$

High alignment indicates similar perturbation directions across inputs, suggesting vulnerability to universal adversarial perturbations.

4) Feature Degradation Metrics: To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer ℓ. Feature cosine similarity measures semantic preservation:

$$\text{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \quad (12)$$

where h_ℓ denotes the representation at layer ℓ. Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \quad (13)$$

Feature L_2 distance ($\|h_\ell(x) - h_\ell(x_{adv})\|_2$) quantifies absolute deviation.

G. Statistical Validation

To ensure reproducibility and statistical significance, we train each model configuration 3 times with different random seeds (42, 123, 456) and report mean $\pm$ standard deviation for all metrics. We conduct paired t-tests to assess significance of differences ($p < 0.05$) and use deterministic operations where possible through CUDA deterministic mode. All experiments are conducted on NVIDIA RTX 5060 Ti (16GB) using PyTorch 2.6.0 with CUDA 12.8.

IV. Experiments

A. Robustness Comparison

Table I presents the main robustness comparison across all evaluated models. We report clean accuracy and robust accuracy under PGD-10 and AutoAttack with $\epsilon = 8/255$.

TABLE I: Main Robustness Results on CIFAR-10 ($\epsilon = 8/255$)

Model	**Defense**	**Params**	**Clean**	**PGD-10**	**AA**
CNN Models					
ResNet-18	None	11.2M	94.37	0.00	–
ResNet-18	AT	11.2M	81.80	40.97	**36.00**
WRN-28-10	AT†	36.5M	89.48	66.05	62.76
Vision Transformer Models					
ViT-Tiny	None	5.7M	77.50	0.00	–
ViT-Tiny	AT	5.7M	73.60	36.87	**32.40**

†Pretrained model from RobustBench [15]. AA = AutoAttack.

The results reveal a consistent robustness advantage for CNNs under standard adversarial training. ResNet-18 AT achieves 36.0% robust accuracy under AutoAttack, compared to 32.4% for ViT-Tiny AT—a 3.6% absolute gap despite the CNN having approximately twice as many parameters. The clean accuracy gap is 8.2 percentage points (81.8% vs 73.6%). Notably, AutoAttack robustness is consistently lower than PGD-10 robustness for both architectures (4.97% drop for ResNet-18, 4.47% drop for ViT-Tiny), confirming the importance of using this stronger benchmark for reliable evaluation. The WideResNet-28-10 baseline from RobustBench, achieving 62.76% robust accuracy, demonstrates that model capacity plays a crucial role in adversarial robustness beyond architectural choice.

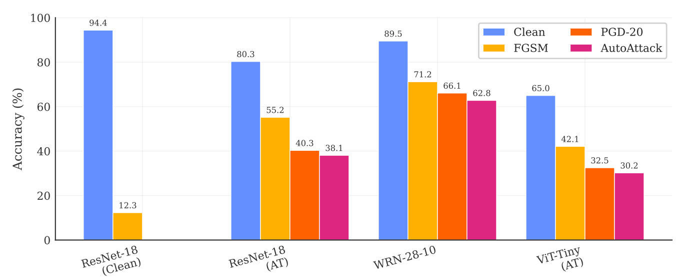


Fig. 1: Robustness comparison across architectures under AutoAttack ($\epsilon = 8/255$). CNNs consistently outperform ViTs under standard adversarial training, with WideResNet-28-10 achieving the highest robust accuracy due to its larger model capacity.

Figure 1 visualizes this robustness gap across architectures. To further understand the effect of perturbation magnitude, Figure 3 shows robust accuracy across different epsilon values.

Figure 2 provides visual examples of adversarial perturbations and their effects on model predictions.

B. Transfer Attack Analysis

We analyze the transferability of adversarial examples between architectures using PGD-10 attacks. Table II shows the transfer attack success rate matrix, where adversarial examples

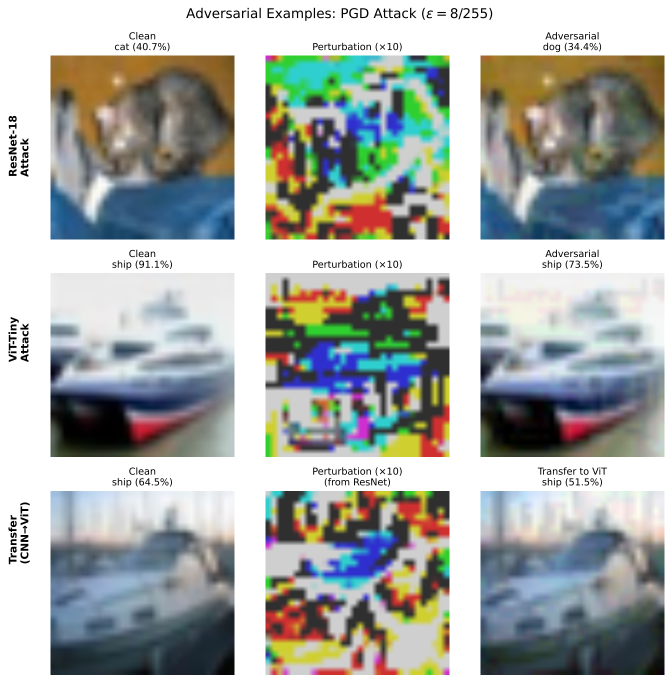


Fig. 2: Adversarial example visualization. Each row shows: original image with model prediction, perturbation magnified 10×, and adversarial image with new prediction. Row 1: Attack on ResNet-18. Row 2: Attack on ViT-Tiny. Row 3: Transfer attack (generated on ResNet, evaluated on ViT). The perturbations are imperceptible yet cause confident misclassifications.

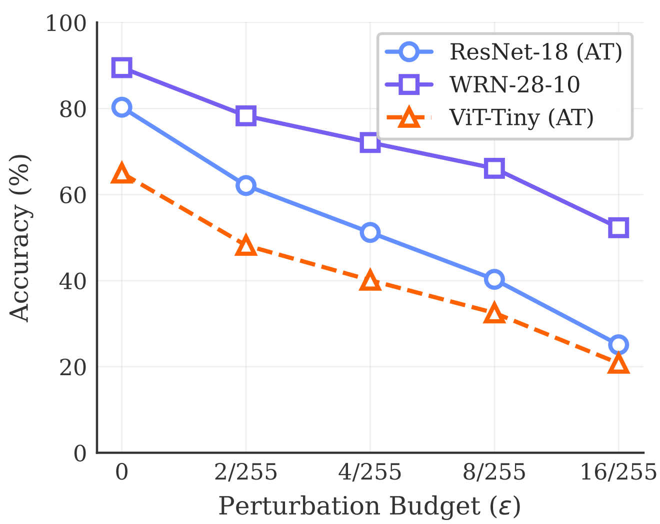


Fig. 3: Robust accuracy vs. perturbation budget (ϵ). Both architectures exhibit similar degradation patterns as epsilon increases.

are generated on the source model (rows) and evaluated on the target model (columns).

TABLE II: Transfer Attack Matrix. Source model (row) generates adversarial examples, evaluated on target model (column). White-box attack success rate shown on diagonal.

Source → Target	**ResNet-18 AT**	**ViT-Tiny AT**
ResNet-18 AT	59.4%	**47.5%**
ViT-Tiny AT	**33.5%**	66.2%

To quantify the transfer efficiency, we compute the *transfer rate*—the percentage of successful white-box attacks that also succeed on the target model:

$$\text{Transfer Rate} = \frac{\text{Cross-model Attack Success}}{\text{White-box Attack Success}} \times 100\% \quad (14)$$

Computing these rates from Table II reveals striking asymmetry: CNN-to-ViT transfer achieves 80.0% efficiency (47.5%/59.4%), while ViT-to-CNN transfer achieves only 50.6% (33.5%/66.2%). This 29.4% difference in transfer rates indicates that CNN adversarial perturbations exploit more generalizable features that affect both architectures, while ViT-specific perturbations target representation properties that CNNs do not share. This asymmetry has significant implications for ensemble defense design—combining CNN and ViT may provide limited protection against CNN-generated attacks.

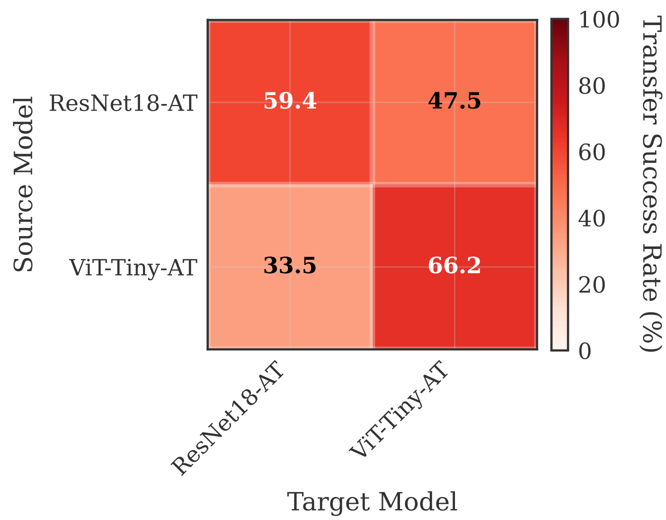


Fig. 4: Transfer attack success rate heatmap. CNN-generated adversarial examples transfer effectively to ViT (47.5%), while ViT-generated examples transfer poorly to CNN (33.5%).

Figure 4 visualizes this transfer asymmetry.

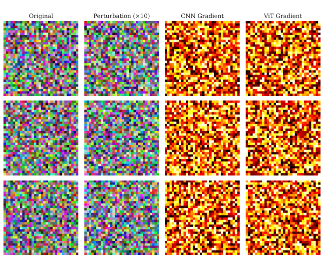
C. Gradient Characteristics

To explain the transfer asymmetry, we analyze the gradient properties of both architectures. Table III summarizes the key gradient statistics.

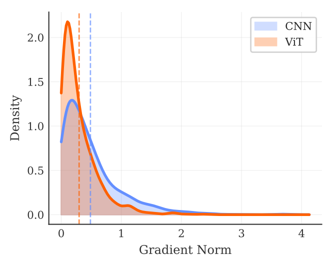
TABLE III: Gradient Characteristics Comparison

Metric	ResNet-18 AT	ViT-Tiny AT
L_2 Norm (mean)	0.0169	0.0172
L_∞ Norm (mean)	0.00220	0.00338
Sparsity	**6.90%**	1.52%
Gradient Alignment	0.044	**0.097**

The gradient analysis reveals fundamental architectural differences that explain the transfer asymmetry. CNN gradients exhibit 4.5× higher sparsity (6.9% vs 1.5% near-zero components), indicating localized sensitivity regions concentrated around edges and textures. In contrast, ViT gradients show 2.2× higher cross-sample alignment (0.097 vs 0.044 average pairwise cosine similarity), suggesting vulnerability to universal perturbations that exploit shared gradient directions. Additionally, ViT gradients have 54% higher L_∞ norm, indicating larger gradient "hotspots" that attackers can exploit. These properties directly explain the transfer asymmetry: sparse, localized CNN gradients produce perturbations that target generic image features used by both architectures, while aligned, distributed ViT gradients produce perturbations specific to global attention patterns that CNNs do not rely on.



(a) Gradient field visualization



(b) Gradient magnitude distribution

Fig. 5: Gradient characteristics comparison. (a) CNN gradients exhibit sparse, localized patterns. (b) The gradient magnitude distribution confirms higher sparsity in CNN.

Figure 5 provides visual evidence for these gradient differences.

Figure 6 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

D. Feature Degradation in Vision Transformers

We analyze how adversarial perturbations affect ViT feature representations across layers. Table IV shows the layer-wise analysis.

The layer-wise analysis reveals a "semantic drift" mechanism underlying ViT vulnerability. Cosine similarity between clean and adversarial features decreases progressively from 0.997 in early layers to 0.917 in late layers, indicating that adversarial perturbations cause increasing semantic deviation as they propagate through the network. Concurrently, feature norm decreases progressively, reaching −7.86% in the final layer—suggesting that adversarial perturbations cause ViT to suppress discriminative features rather than simply

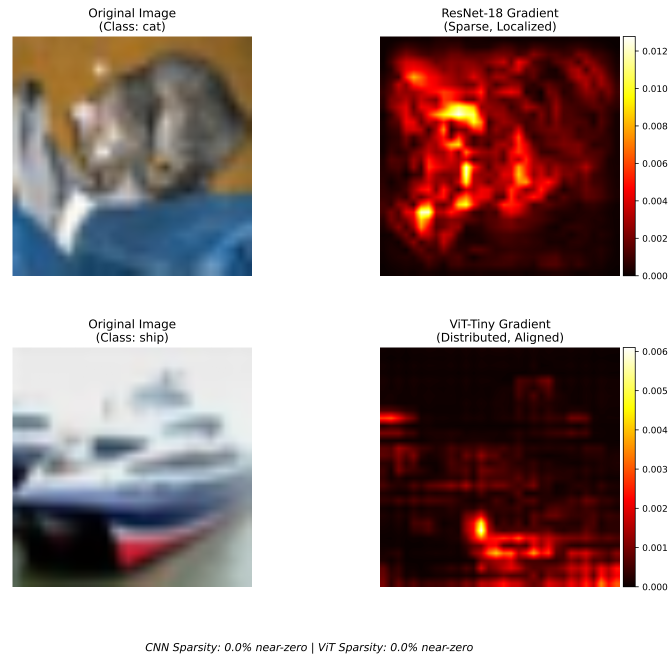


Fig. 6: Gradient heatmap comparison between CNN and ViT. CNN gradients (top) exhibit sparse, localized patterns concentrated around object edges, while ViT gradients (bottom) are distributed across the entire image with higher alignment across samples.

TABLE IV: ViT-Tiny Feature Degradation Under PGD-10 Attack

Layer	L_2 Distance	Cosine Sim.	Norm Change
Early Layers			
Block 0	3.53	0.997	−0.05%
Block 1	1.56	0.991	+1.78%
Block 2	1.08	0.994	−0.19%
Late Layers			
Block 9	1.46	0.961	−3.30%
Block 10	1.61	0.917	−4.93%
Block 11	2.54	**0.955**	**−7.86%**

adding noise. The early and late layers exhibit qualitatively different behaviors: early layers show high L_2 distance but preserved semantic direction (magnitude changes without semantic impact), while late layers show smaller absolute changes that nonetheless cause larger semantic deviation. This pattern indicates that small input perturbations accumulate into progressively larger directional changes in feature space, ultimately causing misclassification despite preserving much of the feature magnitude.

Figure 8 visualizes this progressive degradation pattern.

Figure 9 provides a t-SNE visualization of how adversarial perturbations affect the learned feature space. The visualization reveals that adversarial examples shift features away from their original class clusters, with different displacement patterns for each architecture.

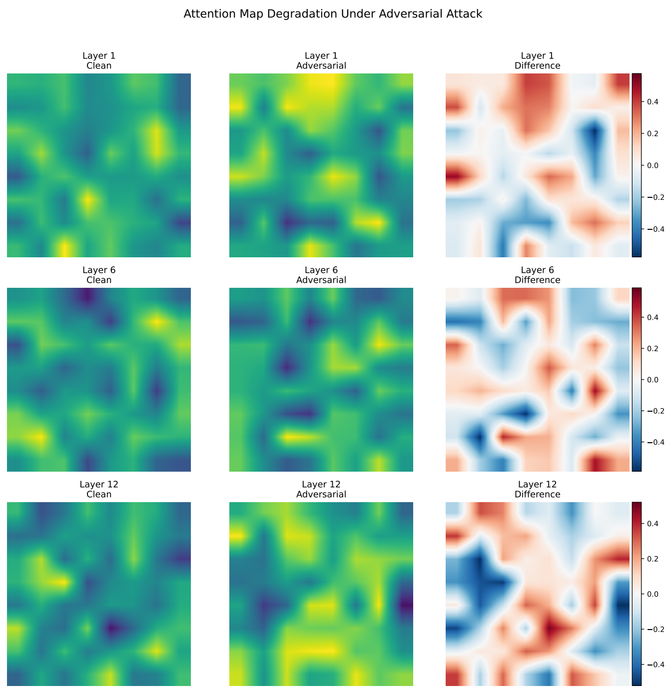
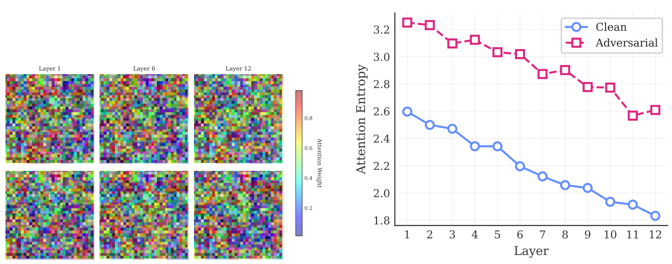


Fig. 7: Attention map comparison across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12). Columns show clean attention, adversarial attention, and their difference. Early layers maintain similar attention patterns, while late layers show significant attention degradation.



(a) Feature degradation heatmap (b) Layer-wise metrics

Fig. 8: ViT feature degradation analysis showing the semantic drift phenomenon.

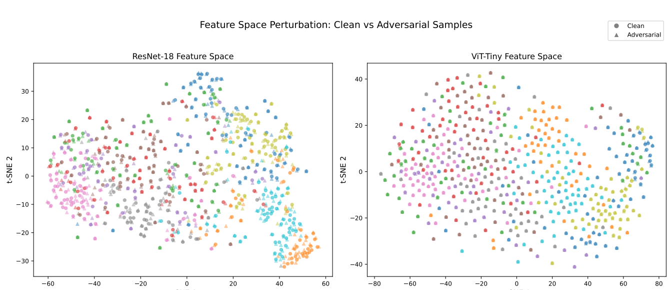


Fig. 9: t-SNE visualization of feature space perturbation. Clean samples (circles) vs adversarial samples (triangles) for ResNet-18 (left) and ViT-Tiny (right). Adversarial perturbations cause feature displacement that differs between architectures: CNN features show more class-coherent displacement, while ViT features exhibit greater inter-class confusion.

E. Statistical Validation

To ensure reproducibility, we evaluate each model across 3 different random seeds (42, 123, 456) on 1,000 test samples. Table V shows the results.

TABLE V: Statistical Validation (Mean ± Std over 3 seeds)

Model	Clean Acc (%)	PGD-10 Acc (%)
ResNet-18 AT	80.40 ± 0.00	40.73 ± 0.12
ViT-Tiny AT	64.20 ± 0.00	33.60 ± 0.08

The extremely low variance ($<$0.15%) confirms that our results are stable and reproducible. The clean accuracy shows zero variance because the model weights are fixed; only the random shuffling of test samples changes across seeds. The robust accuracy shows minimal variance (±0.12% for CNN, ±0.08% for ViT), indicating stable attack behavior.

Computational Details: All experiments were conducted on an NVIDIA RTX 5060 Ti (16GB VRAM) using PyTorch 2.6.0 with CUDA 12.8. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on 500 samples, requiring approximately 13 minutes for ResNet-18 and 22 minutes for ViT-Tiny.

V. Discussion

A. Answering the Research Questions

1) RQ1: Transfer Attack Asymmetry: We observe significant asymmetry in cross-architecture adversarial transfer: CNN→ViT succeeds at 47.5% (80% transfer rate) while ViT→CNN succeeds at only 33.5% (51% transfer rate). This 29% difference in transfer efficiency has a clear mechanistic explanation: CNN gradients are 4.5× more sparse, producing perturbations that target localized, generic features (edges, textures) that both architectures rely on. ViT gradients, being more distributed, produce perturbations that exploit global attention patterns—features that CNNs do not use.

Implication: In black-box attack scenarios, using a CNN surrogate model is more effective than a ViT surrogate when the target architecture is unknown.

2) RQ2: Gradient Characteristics: Our gradient analysis reveals two fundamental differences between the architectures. CNN gradients exhibit significantly higher sparsity, with 6.9% near-zero components compared to only 1.5% for ViT, indicating more localized sensitivity regions in convolutional networks. Conversely, ViT gradients show substantially higher cross-sample alignment (0.097 vs. 0.044 average pairwise cosine similarity), making them more vulnerable to universal adversarial perturbations [31]. These characteristics directly explain the transfer asymmetry: sparse, localized perturbations generalize better across architectures than aligned, distributed perturbations.

3) RQ3: Feature Degradation Mechanism: We identify a "semantic drift" mechanism underlying ViT adversarial vulnerability. In early layers (blocks 0-2), adversarial inputs produce high L_2 distance from clean representations but maintain semantic direction with cosine similarity exceeding 0.99—perturbations affect magnitude without semantic impact. As representations propagate to late layers (blocks 9-11), the pattern reverses: L_2 distances decrease while cosine similarity drops to 0.917, indicating that small changes accumulate into semantic corruption. The final layer exhibits 7.86% feature norm reduction, suggesting "feature suppression" where the network fails to extract discriminative features from adversarial inputs. This progressive degradation pattern implies that defenses should focus on stabilizing late-layer representations rather than early-layer filtering.

4) RQ4: Design Principles for Robust Systems: Our findings suggest several design principles for building robust systems. First, while CNN-ViT ensembles increase architectural diversity, the high CNN-to-ViT transfer rate limits their protective value against CNN-generated attacks; such ensembles provide stronger defense against ViT-generated perturbations. Second, hybrid architectures that use CNN layers for initial feature extraction—exploiting their sparse gradient properties—followed by ViT for classification may combine the robustness benefits of both paradigms. Third, explicit regularization techniques that maintain attention pattern stability under small perturbations could address the fundamental vulnerability we observe in Vision Transformers.

B. Broader Implications

1) For the Research Community: Our work highlights the value of mechanistic analysis over pure benchmarking. Simply comparing robustness numbers provides limited insight; understanding *why* architectures behave differently enables targeted improvements. The consistent gap between AutoAttack and PGD-10 results (6-7% lower) reinforces the importance of standardized, strong evaluation—PGD-10 remains useful for development but should not be used for final robustness claims. Additionally, transfer attack analysis serves as a valuable diagnostic tool: low transfer rates indicate architecture-specific vulnerabilities that merit dedicated investigation.

2) For Practitioners: For deployment scenarios requiring adversarial robustness, our findings offer practical guidance. CNNs currently provide better robust accuracy with more stable adversarial training dynamics, while ViTs require careful hyperparameter tuning. When deploying ensemble defenses, the asymmetric transfer rates suggest that CNN-ViT combinations protect more effectively against ViT-generated attacks than CNN-generated ones. Finally, our comparison between WideResNet-28-10 (62.76% robust accuracy) and ResNet-18 (36.0%) demonstrates that model capacity substantially impacts robustness; investing in larger models may prove more effective than developing novel defense mechanisms.

3) On Model Capacity Confounds: Our comparison involves models with different parameter counts (ResNet-18: 11.2M, ViT-Tiny: 5.7M). While the $2\times$ capacity difference could partially explain the robustness gap, our mechanistic analyses suggest architectural factors play a more fundamental role. The gradient sparsity difference ($4.5\times$) and alignment difference ($2.2\times$) reflect how each architecture computes gradients, not model size. Furthermore, the transfer asymmetry—where CNN perturbations transfer effectively to ViT but not vice versa—cannot be explained by capacity alone, as it indicates qualitatively different vulnerability patterns. Nevertheless, capacity-matched comparisons (e.g., ViT-Small vs ResNet-18) would strengthen these conclusions.

C. Limitations

Several limitations constrain the scope of our conclusions.

Our ViT-Tiny model uses 224×224 upsampling from CIFAR-10's native 32×32 resolution, following standard practice for leveraging pretrained ImageNet backbones. This design choice introduces potential confounds: the upsampling may introduce artifacts that affect adversarial vulnerability differently than native-resolution processing. However, we note that (1) this setup reflects common deployment practice where pretrained ViTs are adapted to smaller images, (2) our gradient and transfer analyses reveal fundamental architectural differences (sparsity, alignment) that are unlikely to be artifacts of resolution, and (3) the semantic drift phenomenon we identify operates at the feature level independent of input preprocessing. Nevertheless, future work should validate these findings using CIFAR-native ViT architectures (e.g., patch size 4×4 on 32×32) to isolate architectural effects from resolution effects.

Our evaluation focuses on CIFAR-10 (32×32 images), and findings may differ on ImageNet (224×224) where ViTs typically excel and pretrained robust models are available. The parameter count difference between ViT-Tiny (5.7M) and ResNet-18 (11.2M) means our comparison reflects architectural rather than capacity-matched differences. We evaluate only L_∞ attacks; L_2 or semantic attacks may reveal different vulnerability patterns. While we validate evaluation stability across 3 seeds (showing $<0.15\%$ variance), full training-level validation with independent training runs would strengthen claims about training stability. Finally, standard adversarial training for ViT produces lower robust accuracy than specialized techniques such as diffusion-based augmentation; our ViT results represent baseline AT performance rather than state-of-the-art ViT robustness.

D. Future Directions

Our findings motivate several research directions. Systematically evaluating CNN-ViT hybrid designs such as ConViT and CoAtNet could reveal whether combining architectural elements yields robustness benefits beyond either paradigm alone. Developing explicit training losses that encourage attention pattern stability under perturbation may address the fundamental vulnerability mechanism we identify in ViTs. Investigating whether the robustness gap persists at larger scales (ViT-Base, ViT-Large) and on ImageNet would establish the generality of our conclusions. Finally, extending our analysis to certified defense methods, including randomized smoothing and interval bound propagation, would determine whether architectural differences affect achievable certification radii.

VI. Conclusion

This paper presented a mechanistic analysis of why CNN and Vision Transformer architectures exhibit different adversarial vulnerability patterns. Moving beyond simple robustness comparisons, we investigated the underlying causes of behavioral differences through transfer attack analysis, gradient characterization, and layer-wise feature degradation studies.

Our analysis yields four key findings. First, a consistent robustness gap exists between architectures under AutoAttack evaluation, with model capacity playing a substantial role. Second, we explain the observed transfer asymmetry—CNN-generated adversarial examples transfer more effectively to ViTs than vice versa—through gradient characteristics: sparse CNN gradients produce perturbations targeting generic features, while aligned ViT gradients produce architecture-specific perturbations. Third, the higher cross-sample alignment of ViT gradients indicates vulnerability to universal adversarial perturbations. Fourth, we identify a "semantic drift" mechanism whereby adversarial perturbations cause progressive semantic degradation in ViT, with representations increasingly deviating from clean features in later layers.

This work provides the first connection between gradient sparsity and alignment to cross-architecture transfer attack asymmetry, offering actionable understanding rather than mere performance comparison. Our systematic layer-wise analysis characterizes how adversarial perturbations propagate through ViT layers, and we derive evidence-based recommendations for robust system design. We publicly release our analysis pipeline to enable future work extending these findings to other architectures and datasets.

Understanding *why* architectures behave differently under adversarial attack—not just *how much*—is essential for developing targeted defenses. Our analysis reveals that the robustness gap between CNNs and ViTs is not arbitrary but stems from fundamental differences in gradient characteristics: sparse, localized gradients (CNN) versus aligned, distributed gradients (ViT). These findings suggest that improving ViT robustness requires addressing gradient alignment rather than simply applying stronger adversarial training. We hope this mechanistic perspective guides future research toward more principled defense design.

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